Experiment 5

Fourier Series Analysis of Periodic Signals

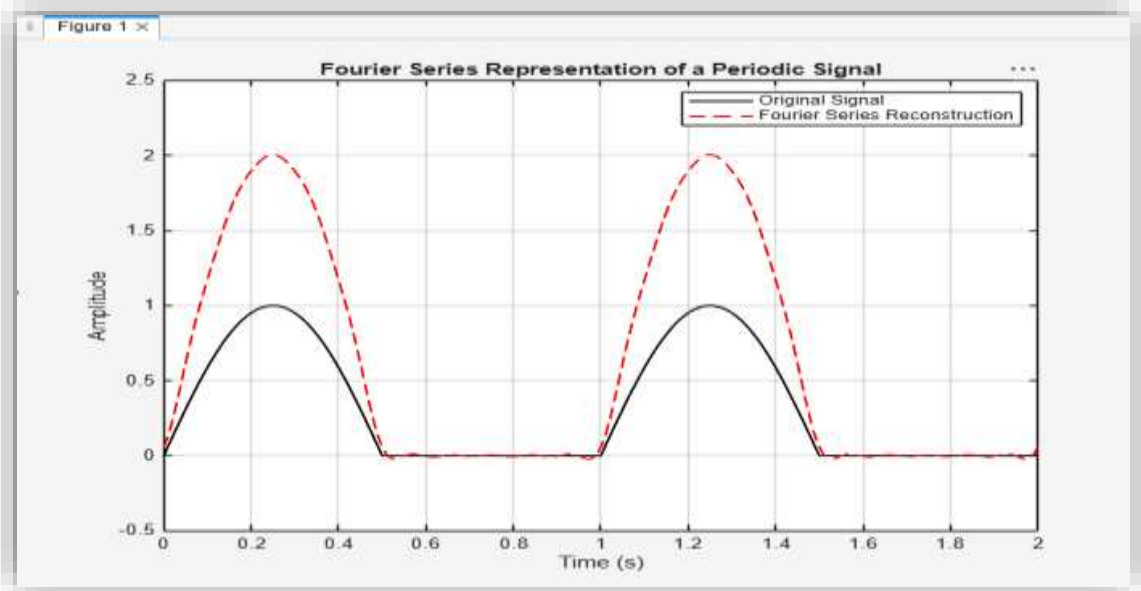
i)

```
/MATLAB Drive/Experiment_5_1.m
clc;
clear;
close all;

figure;

A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
a0 = (2/T)*trapz(t, x)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t, x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t, x.*sin(2*pi*k*f0*t));
end
xFS = a0*ones(size(t));
for k = 1:N
    xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
end
plot(t,x,'k','LineWidth',1.2); hold on;
plot(t,xFS,'r--','LineWidth',1.2);
grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal','Fourier Series Reconstruction');
```

Code



Output